Advances in Therapy® | *Volume 23 No. 4 July/August 2006*

Effects of Angiotensin-Converting Enzyme Inhibitors, Angiotensin II Receptor Blockers, and Their Combination on Microalbuminuria in Normotensive Patients With Type 2 Diabetes

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ABSTRACT

The goal of this study was to compare the effects of lisinopril, losartan, and their combination on microalbuminuria in normotensive patients with type 2 diabetes mellitus. Patients were randomly assigned to 3 groups: group 1 (n=9), group 2 (n=9), and group 3 (n=8) received 10 mg lisinopril, 50 mg losartan, and 10 mg lisinopril plus 50 mg losartan, respectively, each day. For 12 mo, the 24-h urine albumin excretion rate was assessed at 3-mo intervals. At study completion, the urine albumin excretion rate had been reduced significantly in each group (P=.001); however, no significant differences were noted among groups (P=.587). Investigators in the present study have concluded that lisinopril, losartan, and their combination have similar effects on microalbuminuria in normotensive patients with type 2 diabetes mellitus, and that combination therapy does not provide additional benefit.

Keywords: diabetic nephropathy; microalbuminuria; ACE inhibitors; ATII receptor blockers; renoprotection

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0869

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INTRODUCTION

Diabetic nephropathy is the most common cause of end-stage renal failure in developed countries, and initial microalbuminuria as a hallmark of this process is an independent risk factor for the future development of overt nephropathy and cardiovascular disease.[1,2] Therefore, prevention of progression of microalbuminuria to overt proteinuria is crucial in diabetic nephropathy.

Angiotensin-converting enzyme (ACE) inhibitors have been found to prevent progression of nephropathy by blocking the renin-angiotensin system (RAS) through inhibition of angiotensin II (ATII) formation in both types of diabetes mellitus (DM) in hypertensive[3-5] and normotensive[6-8] patients with microalbuminuria. ACE inhibitors were also found to decrease urinary albumin excretion in normotensive, normoalbuminuric patients with type 1 or type 2 DM.[7-9]

The efficacy of ACE inhibitors in renoprotection has led to the development of ATII receptor blockers that block RAS at ATII type 1 receptor levels. In recent studies, it has been found that ATII receptor blockers have beneficial renoprotective effects in patients with type 2 DM who have microalbuminuria[10,11] and overt proteinuria.[12,13]

ACE inhibitors and ATII receptor blockers antagonize the RAS at various levels. Because the objective of treatment is to inhibit ATII effects as completely as possible, these agents may provide a greater reduction in microalbuminuria when used in combination rather than with either drug used alone. Several studies have shown conflicting results for combination therapy in terms of change in proteinuria in patients with DM.[14-19]

In this prospective study, we compared the effects of lisinopril as an ACE inhibitor, losartan as an ATII receptor blocker, and the combination of both drugs on urine albumin excretion rates (UAERs) in normotensive patients with type 2 DM and microalbuminuria.

METHODS

This study was designed as a randomized, prospective study with a duration of 12 mo. Normotensive patients with type 2 DM and microalbuminuria were recruited for the study. Microalbuminuria was defined as UAER of 30 to 300 mg/d in at least 2 consecutive 24-h urine specimens. Patients with type 1 DM, secondary diabetes, chronic hepatic failure, congestive heart failure, chronic renal failure, hypertension (history of taking antihypertensive pills or blood pressure [BP] greater than 130/85 mm Hg), chronic complications of diabetes, malignancy, urinary tract infection, smoking history, and hemoglobin (Hb)A_{1c} values greater than 7% were excluded from the study. All patients gave informed consent to participate in the study in accordance with the Helsinki Declaration, and the study protocol was approved by the local ethical committee.

A total of 34 patients participated in the study. Four patients were lost to follow-up, 2 had adverse reactions to drugs, and 2 had newly diagnosed hypertension that necessitated a change in the treatment protocol. Analyses were performed on 26 patients. Eight patients were only given a diet, 10 were given a diet and oral hypoglycemic agents, and 8 were given a diet and insulin treatment for diabetes management. Protein intake was restricted to 0.8 g/kg/d. Demographic data,

duration of diabetes, body mass index (BMI), whole blood count, serum lipid measurements, hepatic function test results, and measurements of blood urea nitrogen, serum creatinine, and potassium were recorded at the beginning of the study.

Patients were randomly assigned to treatment with lisinopril 10 mg daily (group 1, n=9), losartan 50 mg daily (group 2, n=9), or a combination of lisinopril 10 mg daily plus losartan 50 mg daily (group 3, n=8). Patients were followed at 3-mo intervals for 12 mo. BMI, systolic BP (SBP), diastolic BP (DBP), 24 h UAER, HbA_{1c}, fasting and postprandial plasma glucose values, and serum potassium values were assessed at baseline and at each follow-up visit.

BMI was calculated by dividing the weight in kilograms by the square of height in meters. BP was measured after 5 min rest and after at least 30 min without consumption of tea or coffee. BP measurements were taken with a mercury sphygmomanometer while the patient was sitting up straight and they were repeated 2 min later. The mean of BPs measured 2 min apart was calculated. Mean arterial BP (MABP) was calculated as DBP+1/3 pulse pressure.

Blood samples were collected between 8 and 9 AM after overnight fasting for at least 12 h. Plasma glucose was measured with the glucose-oxidase method (Boehringer-Mannheim, Mannheim, Germany). Liver function tests and blood urea nitrogen, serum creatinine, and potassium measurements were performed with the Hitachi 747 Automatic Analyzer (Boehringer-Mannheim, Mannheim, Germany). Whole blood count was measured with venous blood samples that had been anticoagulated with ethylenediaminetetraacetic acid with an automatic Coulter Counter Model Splus VI (Coulter Electronics, Luton, UK). HbA_{1c} values were measured through latex immunoagglutination with the DCA 2000 Analyzer (Bayer, Leverkusen, Germany). Serum lipid measurements were taken with commercial calorimetric assays (Boehringer-Mannheim, Mannheim, Germany), and 24-h UAER was measured by rate nephelometry with the Beckman IMMAGE® Immunochemistry System and Urine Calibrator (Beckman Coulter Inc, Brea, Calif).

Differences between the clinical characteristics of patient groups at baseline were assessed by χ^2 and analysis of variance, when appropriate. Changes in 24-h UAER, HbA_{1c}, SBP, DBP, MABP, and BMI within and between groups during the study period were assessed by repeated measures analysis of variance. All results were expressed as mean±SD. Statistical analyses were performed with the Statistical Package for the Social Sciences for Windows, release 10.0 (SPSS Inc, Chicago, Ill). $P \leq .05$ was considered statistically significant.

RESULTS

The clinical characteristics of patients at baseline are shown in Table 1. No statistically significant differences were observed among groups with respect to sex, age, length of time since onset of DM, serum lipid measurements, and 24-h UAER, HbA_{1c}, SBP, DBP, MABP, and BMI at baseline.

Table 2 shows changes in 24-h UAER throughout the study. Normoalbuminuria (24-h UAER <30 mg/d) was achieved in 8 patients in group 1, 7 patients in group 2, and 6 patients in group 3 by the end of the study. The reduction in 24-h UAER was statistically significant in all 3 groups at the end of the study (P=.001); however, no statistically significant difference was noted among groups regarding 24-h UAER (P=.587) (Figure). Percentages of reduction in 24-h UAER were 67.6%, 61.3%, and

58.7% for groups 1, 2, and 3, respectively (*P*=.567 for comparison between groups regarding the percentage change in 24-h UAER).

Table 1. Clinical Characteristics of Patients at Study Baseline

	Group 1 (n=9)	Group 2 (n=9)	Group 3 (n=8)
Male/female, n	4/5	4/5	3/5
Age, y	55.1±8.9	55.1±9.2	55.1±9.6
Time since onset of DM, y	7.4±5.5	7.6±5.6	7.6±5.8
Serum cholesterol, mg/dL	256±17	249±23	240±10
Serum triglycerides, mg/dL	265±22	273±18	279±12
24-h UAER, mg/d	70.2±32.9	70.1±16.2	70.1±31.9
HbA_{1c}, %	6.3±0.4	6.0±0.4	6.1±0.3
SBP, mm Hg	120.0±7.5	120.0±9.0	120.0±8.0
DBP, mm Hg	77.8±7.5	78.9±5.5	78.8±3.5
MABP, mm Hg	63.7±10.5	65.2±8.1	65.0±6.5
BMI, kg/m^2	27.6±5.3	27.8±3.5	27.9±5.6

Values are mean±standard deviations.

No differences among groups were statistically significant.

Table 2. Changes in 24-h UAER Throughout the Study

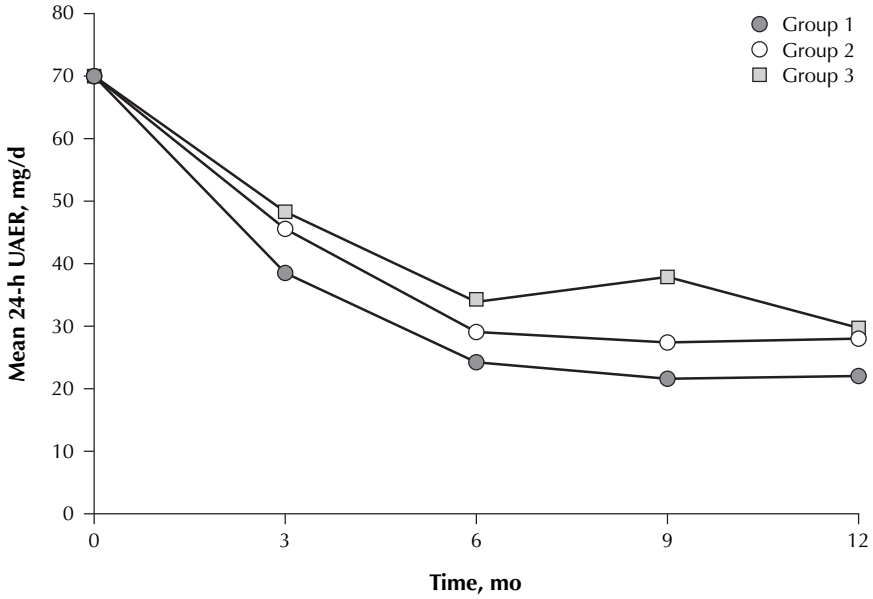
	Group 1	Group 2	Group 3	*P* Value*
24-h UAER, mg/d				
Baseline	70.2±32.9	70.1±16.2	70.1±31.9	1.000
Month 3	38.2±18.7	45.3±18.0	48.1±29.5	.643
Month 6	24.0±12.5	28.9±9.1	33.6±24.2	.482
Month 9	21.5±5.8	27.3±9.5	37.7±32.6	.231
Month 12	21.9±13.8	27.8±14.8	29.6±24.3	.648
ΔUAER, %†	67.6	61.3	58.7	.567
Normoalbuminuria, n	8	7	6	–

**P* values show differences among groups during study periods.

†ΔUAER shows percentage change in 24-h UAER during the study.

Values are mean±standard deviations.

Changes in 24-h UAERs throughout the study.



The most statistically significant reduction in 24-h UAER was observed during the first 3 mo and between the third and sixth months (P=.001 and P=.001, respectively). The reduction between the sixth and twelfth months was not found to be statistically significant (P=1.000).

Statistically significant reductions in SBP, DBP, MABP, and BMI were reported at the end of the study in all groups (P=.001, P=.001, P=.001, and P=.022, respectively, for comparison between baseline and end of study results); however, no statistically significant differences were seen among the groups in reduction in SBP, DBP, MABP, and BMI (P=.967, P=.759, P=.647, and P=.983, respectively). Reductions in SBP, DBP, and MABP were in the normotensive range, and no patients had hypotension throughout the study (data not shown). HbA_{1c} values had not changed significantly by the end of the study (P=.875), and no statistically significant difference was noted among study groups (P=.694). None of the 26 patients reported any adverse reactions to study drugs.

DISCUSSION

In this preliminary prospective study, antiproteinuric efficacies of lisinopril as an ACE inhibitor, losartan as an ATII receptor blocker, and the combination of both drugs were compared. By the end of the study, 24-h UAER had decreased significantly and similarly within each treatment group. Percentage reductions in 24-h UAER were 67.6% in the lisinopril group, 61.3% in the losartan group, and 58.7% in the combination group (P=.567).

Both drugs prevent the progression of nephropathy by blocking RAS at various levels; therefore, it may be reasonable to propose that combination therapy would provide greater benefit than would result from either drug used alone. Although combination therapy did not result in further reduction in 24-h UAER in this study, there was no investigation of whether ACE inhibitors or ATII receptor blockers used alone in higher doses would further decrease 24-h UAER. However, despite significant renoprotection, higher doses of ACE inhibitors might not provide further reduction in 24-h UAER because ATII may be produced through enzymes other than ACE, such as chymase.[15,20] Therefore, ATII receptor antagonists alone or in combination with ACE inhibitors might overcome this problem. However, production of ATII and other angiotensin peptides would be increased with ATII receptor blockers used alone. Because ATII type 1 receptors would be blocked by the drug, ATII and these peptides might activate ATII type 2 receptors and produce adverse effects.[21] This may be another advantage of combination therapy.

Several studies comparing the efficacies of ACE inhibitors and ATII receptor blockers or their combination on UAER in patients with diabetes show conflicting results.[14-19] Differences may be due to variations in study design and patient selection criteria.

Because hypertension and poor glycemic control are confounding factors for UAER, patients with hypertension (BP ≥130/85 mm Hg) and HbA_{1c} greater than 7% were excluded from the study. HbA_{1c} values were not different from baseline at the end of the study in all groups, which shows that glycemic control was maintained throughout the study. None of the patients experienced a change in the diabetes treatment protocol. Another indicator of good glycemic control was the significant reduction in BMI in all groups, showing that patients were compliant with their diet and with treatment throughout the study. Although statistically significant reductions in SBP, DBP, and MABP were reported in all groups, these reductions did not differ between groups and were in the normotensive range. Therefore, reductions in BP were not considered clinically significant.

ACE inhibitors enhance kinase activity, which leads to bradykinin accumulation.[22] Bradykinin reduces glomerular hypertension and hyperfiltration through efferent arteriolar vasodilatation.[22,23] Because clinical studies that compared ACE inhibitors with ATII receptor blockers showed no significant differences in the efficacies of these drugs in terms of change in UAER, it is reasonable to conclude that the renoprotective effects of these drugs are mediated through RAS rather than by kinin metabolism. This may represent an advantage of ATII receptor blockers over ACE inhibitors, in that nonproductive cough seen as an adverse reaction to ACE inhibitors is mediated by bradykinin.

CONCLUSION

This preliminary short-term study shows that ACE inhibitors and ATII receptor blockers have similar efficacy on 24-h UAER, and that combination therapy does not add further benefit. However, long-term studies with greater numbers of patients are needed to elucidate these results.

ACKNOWLEDGMENT

The authors thank Dr. Ergun Karaagaoglu for his assistance with the statistical analysis.

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